

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.



24 Port DO Module: -

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	12 Pin 5.08 mm pitch pluggable screw Terminal Block
Dimensions	157 mm L x 110 mm B x 75 mm H
BPower	Input Power – 15-40 VDC or 24 V AC / DC options Typical – 12V DC @ 150mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Relay Output –

Channels	24
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

Additional Features: -

Communication ports are isolated
 Input power reverse polarity safety
 ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
 EFT IEC 61000-4-4, 50A (5/50ms)
 400V isolation.
 CRC Error check.
 No configuration needed on the IO board

Configuration Settings: -

Communication Speed 9600 – 115200 Kbps (soft settings)
Data Bits 8
Parity None =3
Stop bit 1
CRC Yes
Slave ID Soft Settings selectable (1-247)
Function code 0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address 0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DO 1	00001	0X05,0X0F	RS485	1 Bit Boolean
2	DO 2	00002	0X05,0X0F	RS485	1 Bit Boolean
3	DO 3	00003	0X05,0X0F	RS485	1 Bit Boolean
4	DO 4	00004	0X05,0X0F	RS485	1 Bit Boolean
5	DO 5	00005	0X05,0X0F	RS485	1 Bit Boolean
6	DO 6	00005	0X05,0X0F	RS485	1 Bit Boolean
7	DO 7	00007	0X05,0X0F	RS485	1 Bit Boolean
8	DO 8	00008	0X05,0X0F	RS485	1 Bit Boolean
9	DO 9	00009	0X05,0X0F	RS485	1 Bit Boolean
10	DO 10	00010	0X05,0X0F	RS485	1 Bit Boolean
11	DO 11	00011	0X05,0X0F	RS485	1 Bit Boolean
12	DO 12	00012	0X05,0X0F	RS485	1 Bit Boolean
13	DO 13	00013	0X05,0X0F	RS485	1 Bit Boolean
14	DO 14	00014	0X05,0X0F	RS485	1 Bit Boolean
15	DO 15	00015	0X05,0X0F	RS485	1 Bit Boolean
16	DO 16	00016	0X05,0X0F	RS485	1 Bit Boolean
17	DO 17	00017	0X05,0X0F	RS485	1 Bit Boolean
18	DO 18	00018	0X05,0X0F	RS485	1 Bit Boolean
19	DO 19	00019	0X05,0X0F	RS485	1 Bit Boolean
20	DO 20	00020	0X05,0X0F	RS485	1 Bit Boolean
21	DO 21	00021	0X05,0X0F	RS485	1 Bit Boolean
22	DO 22	00022	0X05,0X0F	RS485	1 Bit Boolean
23	DO 23	00023	0X05,0X0F	RS485	1 Bit Boolean
24	DO 24	00024	0X05,0X0F	RS485	1 Bit Boolean

Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

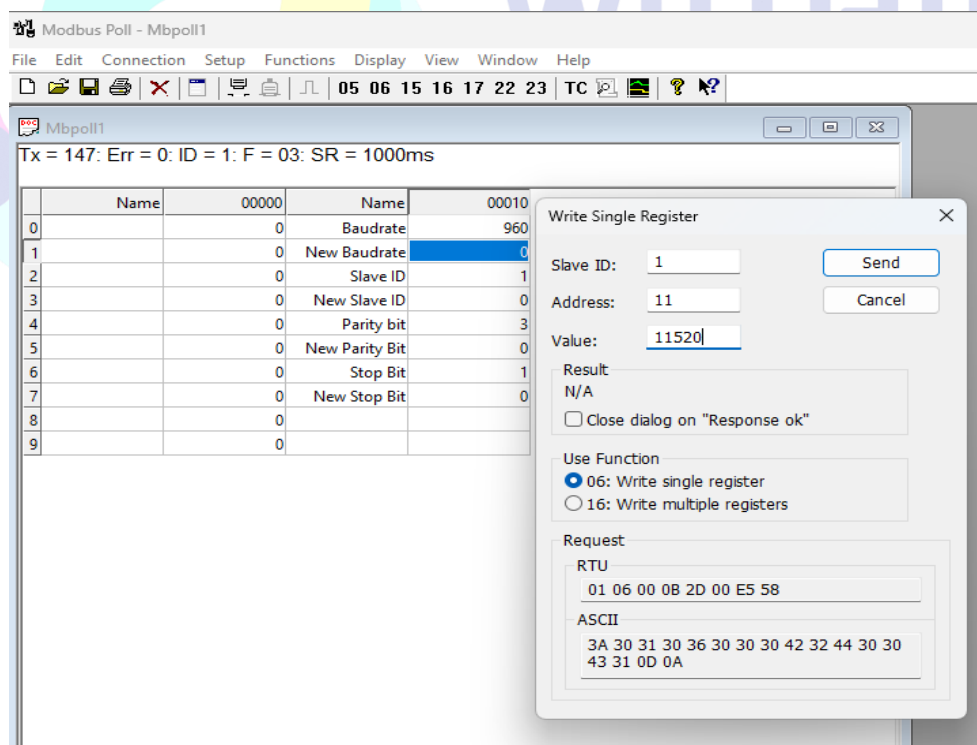
22 AWG Cable	Shielded Twisted pair should be used
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100f

Soft Settings Configuration: -

- Connect The Modbus A and B to cable.

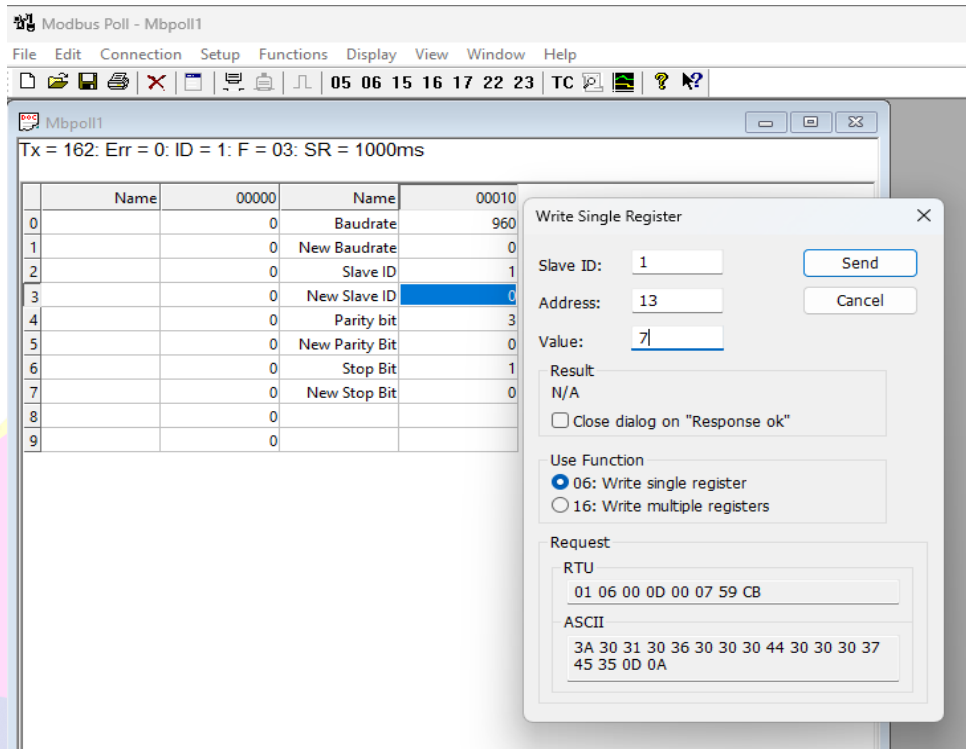
Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default Baudrate is 9600 and slave id is 1).
- After connecting you will get data as shown below.



As you can see in the image above, the resistor 40010 – 960 represents the present baud rate (9600) of the board and 40012 – 1 represents the present slave id of the board.

To change the baud rate and slave id you must enter a value on the 40011 resistors for the baud rate and 40013 for the slave id as shown in the image below.



Hard Reset Setting

If the configured **baud rate** or **slave ID** of the board is unknown or forgotten, follow the procedure below to restore the default settings:

1. Short the **reset jumper** on the board (as shown in the reference image).
2. Reboot the system by either:
 - Power cycling the device, or
 - Pressing and holding the **RESET button** for at least **5 seconds**.
3. After reset, the device will revert to the default communication settings.
4. Default Communication settings :9600 Baud rate , 1 is slave ID

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